

Project Progress Report

Data Warehouse

Bachelor of Data Science, Faculty of Mathematics and Natural Sciences
Universitas Negeri Surabaya

Class	2024INT
Leader	Ketut Shridhara (24031554115)
Member 1	Illona Anindya (24031554222)
Member 2	Nazril Ravi Pratama (24031554129)
Github URL	https://github.com/kiuyha/dwh-socio-economic-etl

General Project Information

Title	Automated Web Scraping-Based Data Warehouse System for X and Reddit about Socio-Economic Discourse in Indonesia
Topic	Socio-Economic
Revision?	Yes/No

Problem Statement Description	Social media platforms, especially X (previously Twitter) and Reddit, have become primary spaces where the public voices their opinions on many fields, one of them is economic conditions. In Indonesia, discussions around fuel price hikes, minimum wage adjustments, exchange rate fluctuations, and mass layoffs generate huge volumes of posts daily, yet this data largely goes unanalyzed in any structured or systematic way [2][5]. The urgency lies in the gap between data availability and analytical accessibility. Policymakers and researchers currently lack tools that continuously monitor and aggregate public sentiment on socio-economic issues at scale [5]. Traditional survey methods are too slow and expensive to capture real-time shifts in public opinion. Meanwhile, the raw data from social media remains unstructured, noisy, and scattered across platforms with incompatible formats, and also increasing unconditionally, so it is making direct analysis impractical without a dedicated pipeline [2][4]. The core challenge is threefold. First, integrating heterogeneous data from two structurally different platforms into a unified, queryable format requires non-trivial schema harmonization [2][4]. Second, the multilingual nature of Indonesian social media, where users freely mix Bahasa Indonesia, slang, abbreviations, English, and local language demands language-aware NLP models rather than general-purpose ones [1]. Third, without a proper data warehouse architecture, any analysis remains ad hoc and non-reproducible [2][4]. This project addresses all three by building a full ETL pipeline backed by a star schema in PostgreSQL, with automated scheduling via Apache Airflow and interactive OLAP exploration through Atoti.
Target Insights to be Achieved	Through the Data Staging process (ETL pipeline) and Data Presentation layer (Atoti OLAP cube), this project targets four main insight directions: 1. Temporal Sentiment Trends: Using the time hierarchy (Year to Quarter to Month to Day) in the CUBE query to track how average public sentiment on specific topics, such as inflation or fuel prices, shifts over

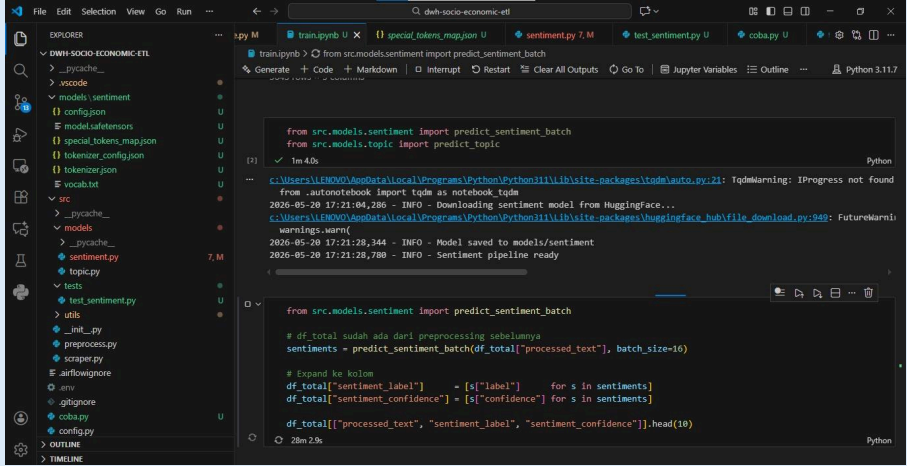
	time [5]. This can reveal whether a particular policy announcement triggered a measurable change in public mood [1][5]. 2. Cross-Platform Sentiment Comparison: Slicing the CUBE along the platform dimension (X vs. Reddit) to quantify whether users on each platform respond differently to the same socio-economic issue and time window [1][2]. The hypothesis is that Reddit discussions tend to be more analytical, while X reactions are more emotionally immediate. 3. Topic Volume Distribution: Using the topic dimension derived from LDA modeling to rank the most-discussed socio-economic issues per quarter [3], so it is revealing whether public attention is event-driven or sustained over time. 4. Engagement and Sentiment Correlation: Dicing the CUBE across the sentiment and engagement dimension to test whether negatively-toned posts consistently attract higher interaction rates than neutral or positive ones [1][5], measured as mean engagement grouped by sentiment label across both platforms [2].
Research References	[1] https://doi.org/10.1007/s13278-023-01030-x [2] https://doi.org/10.1007/s13278-022-00960-2 [3] https://doi.org/10.1007/s10462-023-10471-x [4] https://doi.org/10.1007/s11227-024-06413-1 [5] https://doi.org/10.1007/s13278-022-00922-8

Dataset Details

Dataset Source	Data is acquired through two separate scraping scripts running on a daily schedule via Apache Airflow. For X (Twitter), a custom Nitter-based scraper collects posts targeting keywords such as inflasi, BBM, UMR, kurs rupiah, harga pangan, and PHK. Nitter.net is a free, open-source alternative front-end for X that does not need to log in and does not use any JavaScript to load the content, making it easier to scrape at a massive scale compared to scrapers like TweetHarvest, which need a token from a real account, which might be biased to the account's preferences. For Reddit, data is pulled via api.pullpush.io from r/indonesia, r/economy, r/investasi, and r/personalfinance. Each daily run stores results as raw JSON files on the server before entering the staging pipeline.
General Dataset Description	The dataset consists of public social media posts in Indonesian covering socio-economic discourse from 2025 to 2026, with backfill extending to 2023. Each X record contains a tweet ID, post text, timestamp, and engagement metrics including likes, retweets, and replies. Each Reddit record contains a post ID, title, body text, UTC timestamp, score, upvote ratio, and comment count. Both sources are unified into a single schema during transformation.
Special Characteristics of the Dataset	The text is highly sparse. Indonesian social media writing mixes formal terms with slang, abbreviations, and code-switching between Indonesian and English, which produces a scattered vocabulary space relative to document length. Sparsity in the document-term matrix is expected to exceed 95% after vectorization. Sentiment distribution is likely imbalanced toward negative. Topics like fuel price increases and layoffs naturally generate more negative reactions, with a preliminary estimated split of roughly 55% negative, 30% neutral, and 15% positive, to be confirmed after full extraction.

	Engagement metrics follow a power-law pattern. Most posts receive near-zero interaction, while a small number attract disproportionate attention. High-engagement posts are retained and flagged with an engagement tier label rather than treated as outliers, since they carry analytical weight for the correlation analysis planned in the OLAP layer.
--	--

Data Staging Progress

Extraction Process	Scraping scripts for X (via Nitter) and Reddit (via api.pullpush.io) are fully operational. Data is stored directly into Supabase instead of JSON files to prevent data loss in case of server interruptions. Scraping has been completed for the years 2023, 2024, 2025, and 2026. Search queries used are documented in the <code>app_config</code> table in Supabase. Collected data is stored in 2 tables: <code>raw_tweets</code> and <code>raw_reddit</code> . The tables are partitioned by quarter from Q1 2023 to Q4 2026 to support efficient range-based querying.
Transformation Process	The preprocessing script is complete. For sentiment analysis, the model <code>mdhugol/indonesia-bert-sentiment-classification</code> from HuggingFace was selected as the backbone, which is a fine-tuned IndoBERT model trained specifically for Indonesian text, producing 3 labels: positive, neutral, and negative. The model is loaded once at module level to avoid redundant reloading across pipeline calls, and is cached locally to <code>models/sentiment/</code> after the first download to eliminate internet dependency on subsequent runs. The <code>predict_sentiment_batch</code> function processes text in batches with input sanitization and truncation at 512 tokens, outputting a sentiment label and confidence score per post. Unit tests covering positive/negative cases, output structure, edge cases (empty/None inputs), and batch consistency have been written and validated using <code>pytest</code> across 5 test cases. Initial inference results show confidence scores consistently above 0.9, indicating strong model certainty.  Figure 1. Sentiment Model Inference Output: IndoBERT Pipeline (<code>mdhugol/indonesia-bert-sentiment-classification</code>) Transformed results including <code>sentiment_label</code> and <code>sentiment_confidence</code> columns will be stored in the <code>staging_transformed</code> table. Full pipeline execution is pending completion of the topic modeling script currently in development.
Load Process	The load script is complete and designed to read from <code>staging_transformed</code> and convert the data into the OLAP star schema.

The full schema is documented in `schema.sql` and the entry point is `main.py`. Airflow integration supports manual runs with configurable year targeting and optional scraping skip. Automated runs are scheduled to trigger scraping annually. End-to-end execution has not yet been tested as it is pending completion of the transformation stage.

- **Database Size:** As of the current extraction stage, the `raw_tweets` table contains 5,554 records and the `raw_reddit` table contains 89 records, the data are from 2023 to 2026 across all configured socio-economic keywords.
- **PostgreSQL Technologies Used:** `PARTITION BY RANGE` has been applied to `raw_tweets` and `raw_reddit`, partitioned per quarter from Q1 2023 to Q4 2026. Additional features including `Materialized View`, `Index`, and `pg_trgm` extension have already been configured in Supabase. But these features have not yet produced measurable impact because the OLAP star schema has not been populated with transformed data.
- **Performance Benchmarking:** Not yet conducted because the OLAP schema is still empty and the full ETL pipeline has not been executed end-to-end. At the current stage, only preliminary database performance during extraction is observable, showing stable query execution and partitioned table operation in Supabase. Detailed benchmarking results involving `Materialized View`, `Index`, and `pg_trgm` optimization will be evaluated after the load process successfully populates the star schema.

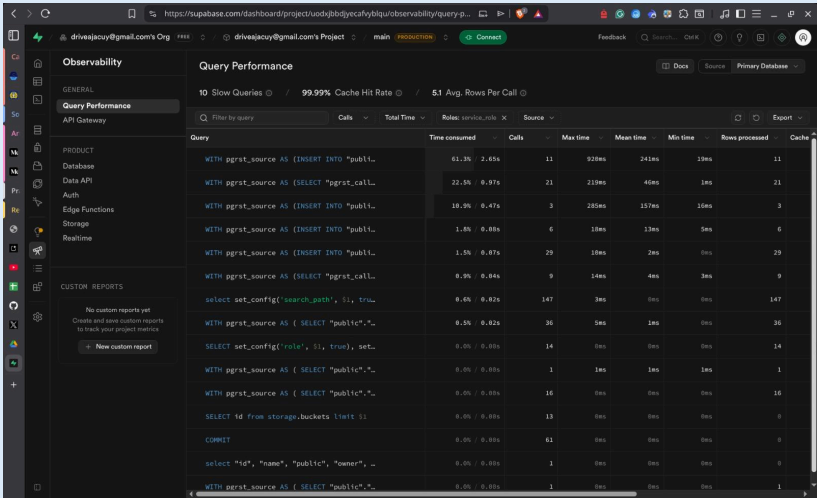
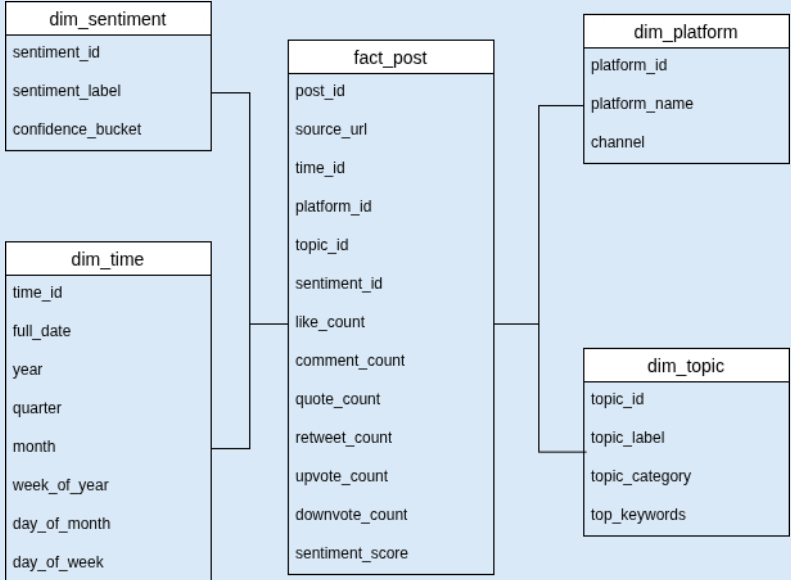


Figure 2. Supabase Query Performance During Extraction Process

Data Presentation Progress

Subject Focus	The OLAP analysis focuses on public sentiment toward socio-economic issues in Indonesia across two platforms over time. The subject connects three dimensions: when something was posted, where it was posted, and what issue it addressed. The goal is to let analysts move between broad patterns, for example, national sentiment in Q1 2025, and specific slices, for example, Reddit users discussing BBM in a particular week, without writing raw SQL each time.
---------------	--

Star Schema	The schema uses one fact table and four dimension tables. 1. fact_post: Acts as the central fact table containing the primary measures for the OLAP cube, including like count, comment count, quote count, retweet count, upvote count, downvote count, and sentiment score. 2. dim_time: Acts as a dimension supporting time-based roll-ups. It defines a time Hierarchy (Year > Quarter > Month > Week > Day) enabling drill-down analysis from annual trends to day-level views. 3. dim_platform: Acts as a dimension to filter by data source. It defines a structural Hierarchy (Platform > Channel/Subreddit), enabling direct engagement comparison between the X and Reddit communities. 4. dim_topic: Acts as a dimension storing the socio-economic categories. It allows OLAP slicing and dicing based on the final topic labels (e.g., inflasi, UMR). 5. dim_sentiment: Acts as a dimension storing the final classification labels (positive, negative, neutral) and confidence buckets (High, Medium, Low), allowing the Atoti DataMart to aggregate measures by public mood.  <pre> graph LR fact_post[fact_post] --- dim_sentiment[dim_sentiment] fact_post --- dim_time[dim_time] fact_post --- dim_platform[dim_platform] fact_post --- dim_topic[dim_topic] </pre> Figure 3. Star Schema
OLAP Result	Not yet conducted. OLAP querying will be conducted once the load process is complete and data is fully populated into the star schema.
OLAP Visualization	Not yet conducted. Atoti DataMart and Widget implementation will follow once the star schema is populated and OLAP queries are validated.